

Setup Guide: Ramp Climber

Time Allocation: 90 Minutes

Challenge Overview

The robot must climb an inclined wooden board set at a 15-degree angle and hold its position at the top.

Environment & Arena Setup

- **Ramp Construction:** Securely mount a smooth wooden board (approximately 2 to 4 feet long and at least 1.5 times the width of the standard student robots) at a strict **15-degree angle** against a stable elevated platform or wall.
- **Surface Check:** Ensure the wooden board is clean, free of major debris or oily residue, and provides a consistent friction surface across all testing lanes.
- **Boundary Markers:** Use painter's tape to clearly define the **starting zone** at the base of the ramp and the **success zone** (holding area) at the very top of the ramp.
- **Safety Measures:** Secure the top and bottom of the ramp structure with weights or clamps so it does not wobble, slide, or tip over when heavy robots drive onto it.
- **Testing Station:** Set up a queue area nearby so student teams can test their robots one at a time without crowding the incline arena.